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Instability Measurements on a Cone-Slice-Flap in Mach-6 Quiet Flow

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Reattachment behavior and shear-layer instabilities are investigated on a 7° cone, featuring a slice parallel to its axis and a 20° compression ramp. Experiments and computations were conducted as part of the NATO STO Research Task Group AVT-346, which is focused on hypersonic boundary-layer studies on complex geometries. Experiments were performed in the Boeing/AFOSR Mach-6 Quiet tunnel at a freestream unit Reynolds number of $2.2{\times}10^6$/m. Accompanying DNS computations were performed at the same conditions using the open-source suite OpenFoam. Mean-flow schlieren visualization in quiet conditions reveals ramp-induced flow separation at the slice leading edge, and reattachment on the ramp. Heat transfer measurements corroborate these findings with increased heat transfer on the ramp near reattachment. In addition, two streamwise streaks of high heating originate on the ramp edges. High-framerate schlieren images captured at 100,000 Hz reveal low-frequency oscillations indicative of bubble breathing at very low frequencies (700–1100 Hz) which corroborates findings from surface pressure measurements which, in turn, are consistent with the DNS results. Mean-flow and time-accurate solutions captured in the simulations show good qualitative comparison with experimental results, unsteadiness in the reattachment zone is observed and quantified and the topology of three-dimensional separation on the slice is discussed.

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I. Introduction

A critical challenge in advancing hypersonic technologies is the understanding of flow separation and boundary-layer transition. Regions where a boundary-layer on the surface of a high-speed vehicle transitions from laminar to turbulent and reattaches further downstream is accompanied by a significant change in heat transfer (aerodynamic heating), aerodynamic moments, and stability [1–3]. Regions of separated flow present on any maneuvering hypersonic vehicle with three-dimensional control surfaces further complicate the prediction of vehicle-level performance. Aerothermal loads and instabilities on control surfaces are largely impacted by both the state of the incoming boundary layer [4] as well as phenomena downstream near reattachment impacting upstream stability [5]. The compression corner formed between the vehicle body and a deflected control surface causes the boundary layer upstream to be subject to an adverse pressure gradient. As the deflection angle increases, the flow in the boundary layer is slowed, eventually resulting in near-wall flow reversal at a large enough pressure gradient. This reversed flow causes shear layer separation near the

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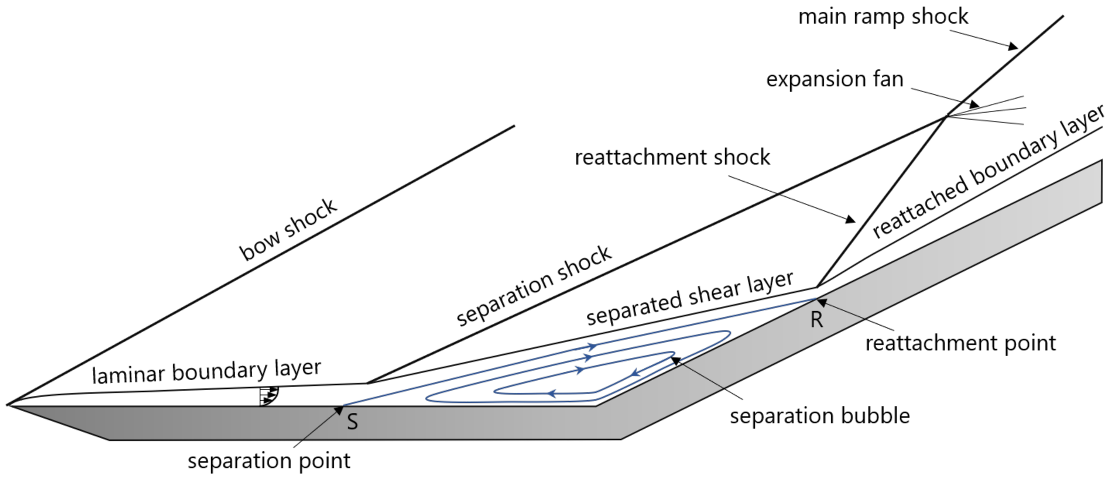

Fig. 1 Shock wave/boundary-layer interaction over a compression corner.

corner. An example of separated flow on a compression corner is shown in Fig. 1 where "S" is the region of separation, and "R" is the region of reattachment [6]. The point at which the boundary layer reattaches can be accompanied by significant surface heating and pressure loads. In real flows, these regions vary in position and intensity with time. Flight tests like the AMaRV [7] and the NASA space shuttle reentry [8] demonstrate the criticality of control surfaces if a vehicle is to fly any trajectory other than ballistic. In addition, the X-33, X-38, and X-43 all had significant errors in their prediction of heating and transition around control surfaces when compared to flight data [9–11].

Cone-slice-ramps are geometries consisting of a sharp or spherically blunted cone with a slice parallel to the cones axis and a ramp positioned on the slice such that it forms a compression corner [12]. In 2021, McKiernan experimented with cone-slice-ramp bodies at 0° AoA in Mach-6 flow with atmospheric-level freestream disturbances [6]. He captured surface pressure and heat transfer measurements along with oil flow visualization to aid in visualization of the separation bubble footprint on a 7° cone with several ramp angles. No off-surface visualization tools were used for these experiments, so off-surface shear-layer fluctuations and mean-flow separation structures were not measured. Symmetric streaks of high Stanton number beginning on the side-edges of the ramp just downstream of the leading edge manifested on the surface and increased in amplitude with an increase in Reynolds number. Computations conducted on the same model geometry by Thome et al. in 2018 found similar streaks of heating on the ramp [13]. They postulated that these streaks were a result of a pair of counter-rotating laminar vortices originating from edge effects, rather than from a transitional or turbulent boundary layer. In addition, they demonstrated the significance of small yaw angles on these vortices and on surface heat flux. Additional computations by Quintanilha and Theofilis at the University of Liverpool in 2021 found similar edge-effects in their steady mean-flow surface pressure and heat transfer for a low-Reynolds-number laminar separation ($Re_\infty = 2\times10^6$) [14]. They presented a contour of zero wall-shear stress as the separation bubble footprint. In 2022, Pandey et al. captured schlieren on a similar cone-slice-ramp geometry at Mach 8, and noted significant unsteadiness and coupling between the separated shear layer and separation shock [15]. Very-low-frequency unsteadiness indicative of large-scale separation bubble breathing was observed to couple strongly with the motion of the separation shock (1.4 kHz). This paper presents ongoing experimental and computational efforts to study separation unsteadiness and reattachment heating patterns on a cone-slice-ramp in quiet hypersonic flow.

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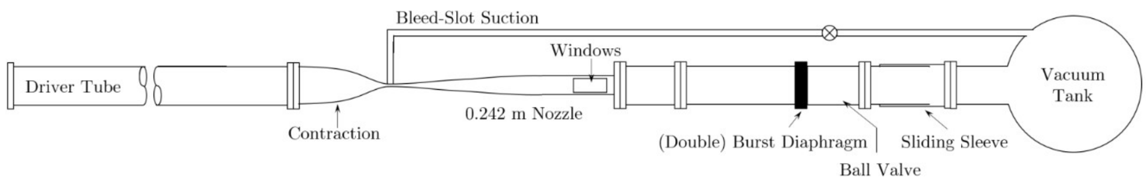

Fig. 2 Schematic of the Boeing/AFOSR Mach-6 Quiet Tunnel.

II. Experimental Apparatus

A. Boeing/AFOSR Mach-6 Quiet Tunnel

The Purdue University Boeing/AFOSR Mach-6 Quiet Tunnel Ludwieg tube was the experimental facility used for this test campaign. A high-level schematic of the facility is shown in Fig. 2. A unique feature of the BAM6QT is that it can operate with very low freestream noise levels. Flight-test data captured at Mach 2 near 11 km altitude show that freestream pressure fluctuations vary between 0.0199% and 0.0057% [16] of the mean pitot pressure, whereas in conventional hypersonic tunnels, noise emanating from the turbulent boundary layers can be several orders of magnitude higher than this [17]. When operating in its low-noise configuration, the BAM6QT has been characterized to have freestream pressure fluctuations of less than 0.02% of the mean pitot pressure in a test rhombus more than 1 m in length [18, 19]. This facilitates more true-to-flight boundary-layer transition studies than would otherwise be available in a noisy tunnel.

The BAM6QT can operate with initial stagnation pressures ranging from 5–300 psia and a typical initial stagnation temperature of ~430 K. Total pressures to achieve the desired Reynolds numbers were ~25 psia. Test times are on the order of 4–6 seconds.

B. Model and Instrumentation

The cone-slice-ramp model used for this experiment and is shown in Fig. 3. The model is composed of an aluminum 6061-T6 7° half-angle cone with a single slice cut parallel to the axis of the cone. The ramp installed 75 mm downstream of the leading edge of the slice induced a flow turning angle of 20° to the slice. This angle was chosen to characterize the separation bubble while still facilitating full reattachment on the ramp. The ramps are 0.091 m in length and 0.051 m in width. The nosetip was machined from 17-4 stainless steel with a radius of 110 µm. A precision angle of attack adapter designed by Chynoweth was used to achieve a 0.0° angle of attack [20]. Studies have been conducted by Francis et al. at additional angles of attack, ramp angles, and Reynolds numbers up to $Re_\infty = 12.0\times10^6$ in the BAM6QT [21].

Positions of surface-mounted sensors on the conical region and slice are shown in Table 1. Sensors are organized by the axial position from the nosetip and azimuthal position from the top centerline. Four PCB-132B38 sensors mounted on the cone at the same axial station at 90° offset angles are used to estimate small variations in angle of attack and sideslip angle.

Table 1 Azimuthal angle of attack/yaw sensor locations on cone for the axis-aligned slice model.

Sensor Type	Axial Location (m from nosetip)	Azimuthal Location (° from top dead center)
PCB	0.245	0°
PCB	0.245	90°
PCB	0.245	180°
PCB	0.245	270°

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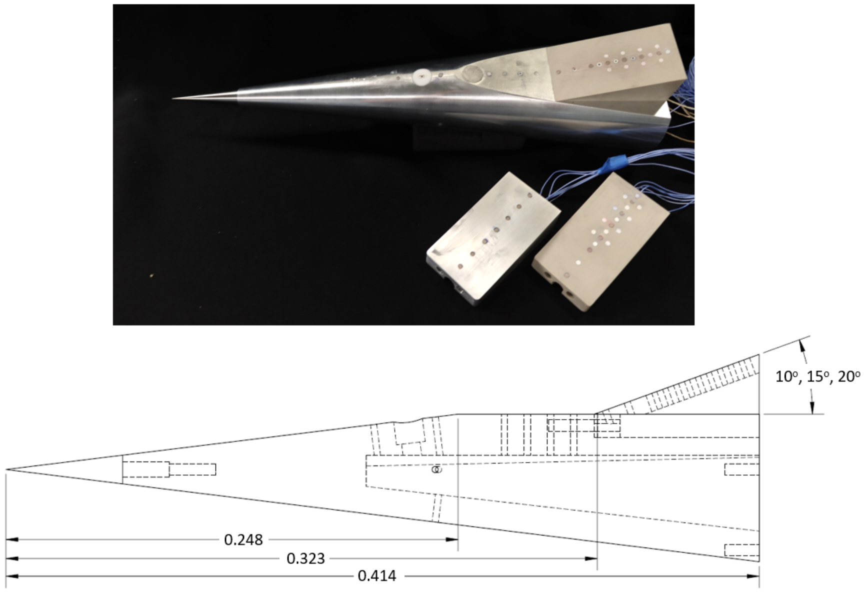

Fig. 3 Photograph and schematic of cone-slice-ramp model investigated in BAM6QT.

1. Finite-Span Compression Ramps

Ramps were fabricated from Poly-Ether Ether Ketone Plastic (PEEK) to facilitate IR heat transfer measurements without the use of temperature-sensitive paint [22]. PEEK is a high-emissivity, easily-machinable plastic that makes it ideal for heat transfer measurements on complex geometries. Following installation of the ramps, the compression angle between the slice and the ramp was measured to ensure that there was no more than a ±0.05°deviation from the nominal ramp angle using a Pro 3600 digital protractor from SmartTool Technologies with 0.01° resolution. An engineering drawing of the 20° ramp is shown in Fig. 4, with sensors labeled in coordinates relative to the model nosetip.

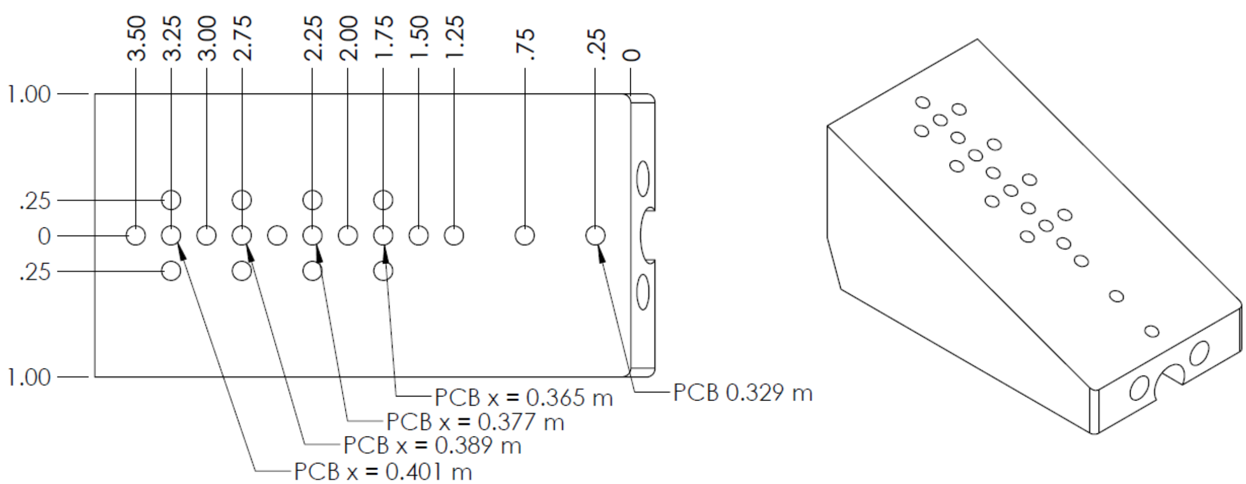

Fig. 4 Detailed schematic the 20° ramp. All units in inches unless noted otherwise.

The axial positions of the sensors relative to the model leading edge and their types are listed in Table 2. Since significant off-centerline measurements have already been captured on similar geometries at 0° angle of attack, and previous studies have already captured trends of spanwise effects on reattachment pressure fluctuations [6, 23, 24], sensors were installed exclusively on the centerline of the model for this set of experiments. Some ramps were designed with no instrumentation ports to capture heat transfer on more of the surface.

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Table 2 Sensor types and locations on the 20° ramp.

Sensor Type	Axial Location (m from nosetip)	Spanwise Location (m from centerline)
Kulite	0.329	0.0
Kulite	0.342	0.0
Kulite	0.365	0.0
Kulite	0.377	0.0
Kulite	0.389	0.0
Kulite	0.401	0.0

2. Surface Pressure

Pressure-fluctuation measurements were made using surface-mounted Kulite XCE-062-15A sensors installed along the center-line of the model slice and ramp. To avoid signal aliasing, Kulite DC and AC signals were sampled at 500 kHz and 2 MHz, respectively, and low-pass filtered at 200 kHz due to their resonance peak at around 300 kHz. The Kulites facilitate measurement of frequency content below 11 kHz, which contains expected SWLBI frequency regimes. Initial analyses use power spectral densities (PSD) computed from Kulite traces to determine where and at what frequencies instabilities exist. PSDs were calculated using Welch's method with a 50% overlap and 100-Hz frequency resolution over a sample time of 0.1 seconds. Amplitudes were normalized with a tangent-wedge approximation of the surface pressure.

Since the surface-mounted Kulites measure exclusively sub-atmospheric static pressures, they were calibrated during each entry by emptying the tunnel to vacuum in known pressure intervals and measuring their voltage at each interval. However, while they have a repeatable sensitivity, there is an inherent temperature effect on the offsets of Kulites that is difficult to characterize in the BAM6QT. Since the model has several minutes to cool between conducting an experiment and a low-pressure calibration, calibrations could not be captured at representative temperatures that are achieved during an experiment. Thus, in absence of a proper calibration method, the static pressures reported may have an incorrect offset.

C. High-Speed Schlieren Imaging

A traditional Z-type schlieren imaging system was used for off-surface flow-field visualization. A CAVILUX Smart laser was used as the light source, producing monochromatic 630-nm light pulses with 10-ns duration continuously at 100 kHz. A Phantom TMX 7510 high-speed camera to record high-speed video of the flow. The camera has a maximum frame rate of 1.75 MHz (based on the resolution, reduced at higher resolutions) and a speed of about 0.25 µs. Frames were captured at 100,000 Hz, resulting in 10 µs between frames. This frame rate was chosen to maximize the viewing window while maintaining a Nyquist limit of greater than 50% above expected separation bubble fluctuation frequencies and mitigate signal aliasing for frequency spectra analysis. Mean-flow schlieren was computed as an average of 5000 frames, equivalent to 0.05 seconds. Given that traditional schlieren is an inherently integrative process, the impact of density gradients on refracted light captured by the camera is collapsed onto a single plane, precluding the investigation of three-dimensional effects.

To maximize the schlieren field of view, flat 5.5×8.375 inch optical-grade sapphire windows were installed for schlieren imaging. The window pressure rating limited the effective maximum stagnation pressure to 170 psia for this test campaign, equivalent to unit $Re_\infty = 12.0\times10^6/m$. This schlieren configuration has been used in BAM6QT to measure transient fluid effects including boundary layer instability growth[25–27], recirculation bubble unsteadiness [28–30], and hypersonic inlet shock wave-boundary layer interaction [31].

D. IR Thermography

Infrared (IR) thermography visualization of the ramp surface was used to reveal the severity of reattachment heating. The techniques used to measure surface heat transfer in this study were originally developed by Zaccara et al. [32] and Edelman [22]. An Infratec ImageIR 8300 hp camera was used for IR thermal imaging of ramp surfaces. The Infratec has a pixel resolution of 640×512, a 0.02 K temperature resolution, an accuracy of ±1 K, and a maximum frame rate of 355 Hz. The camera sensor measures mid-IR wavelengths between 2.0 and 5.7 µm. Images in this study were captured at 100 Hz. A 12 mm, f/3.0 wide-angle lens was used to visualize the heat transfer on the full ramp surface in a single run. IR-transparent silicon windows were used for IR imaging.

III. Direct Numerical Simulations

Experimental work is supported by DNS work to uncover the underlying physical mechanism(s) responsible for flow unsteadiness. Presently, preliminary numerical studies have been carried out at a single free-stream unit Reynolds number of $Re_\infty = 2.2{\times}10^6/m$, corresponding to the experimental setup. A combination of time-accurate solutions and residuals algorithm (RA) [33] is employed aimed at recovering the least-stable global instability at the chosen set of conditions. Further analysis of temporarily saturated flow is performed to probe flow unsteadiness.

A. Computational Setup

The geometry used to create the computational domain is shown in Fig. 5 and is identical in dimensions with the Purdue model with 20° ramp. By the virtue of angle-of-attack and yaw-angle being both zero, computational resources are drastically reduced by simulating only half of the domain with a symmetry boundary condition imposed along the center-line of the model. It should be noted that such choice of boundary conditions does not permit existence of anti-symmetric modes and therefore can affect onset of global instabilities and subsequent nonlinear interactions. Here, special treatment has been applied to the sharp corners and the nose of the geometry. The former have been smoothed with 2-mm fillets to reduce numerical errors introduced to the solution, while the latter has not been resolved and has been treated as a singular point instead. Sutherland's law for air is used to calculate the dynamic viscosity as a function of temperature. For all the computations, the geometry is constrained with an isothermal boundary condition at 300K and all flow conditions are shown in Table 3.

Table 3 Flow conditions for the comutational study.

u_∞ [m/s]	p_∞ [Pa]	T_∞ [K]	T_{wall} [K]	Re_∞	Ma_∞
865.37	126.48	51.76	300.0	2.20×10^6	6.0

Here, hexahedral (mesh 1) and polyhedral (mesh 2) grids with prism inflation layers have been used for the investigation. The hexahedral grid has been generated by an open-source cfMesh* library for automatic mesh generation that is fully integrated within the OpenFoam† package. Mesh 2, on the other hand, was generated with industry-standard ANSA software‡ and subsequently treated by OpenFoam built-in functionality to obtain polyhedral structure of the grid. Temporally resolved signal for further analysis is collected at three probe locations: P1 at (0.2025, 0.0255, 0), P2 at (0.325, 0.032, 0) and P3 at (0.4, 0.06, 0), where z = 0 coincides with the symmetry boundary conditions. The location of the probes are shown in Fig. 5.

This investigation is carried out by means of an open-source solver, rhoCentralFoam [34], which is a part of the OpenFoam suite. Solutions presented herein have been computed in a time-accurate manner using first order implicit Euler and second-order implicit Crank-Nicolson time discretisation schemes on meshes 1 and 2 respectively. Time-accuracy has been ensured by adaptive time-step based on the maximum unsteady CFL number of 0.6, for grid 1, and 0.8, for grid 2, at all times. Space discretisation has been achieved with a central, second-order, Riemann-free scheme by Kurganov and Tadmor [35] that uses MUSCL reconstruction [36]. Table 4 summarises the differences between the two computational strategies.

Table 4 Summary of the computational setups.

	Number of cells	Cell shape	Mesh generation	Temporal scheme	Max. CFL number
Mesh 1	9.79×10^6	hexahedral	cfMesh	implicit Euler	0.6
Mesh 2	10.35×10^6	polyhedral	ANSA	implicit Crank-Nicolson	0.8

*Visit `https://cfmesh.com` for further information about the cfMesh library.
†Visit `https://www.openfoam.com/` for further information about the OpenFoam suite.
‡Visit `https://www.beta-cae.com/ansa.htm` and further information about the ANSA meshing software.

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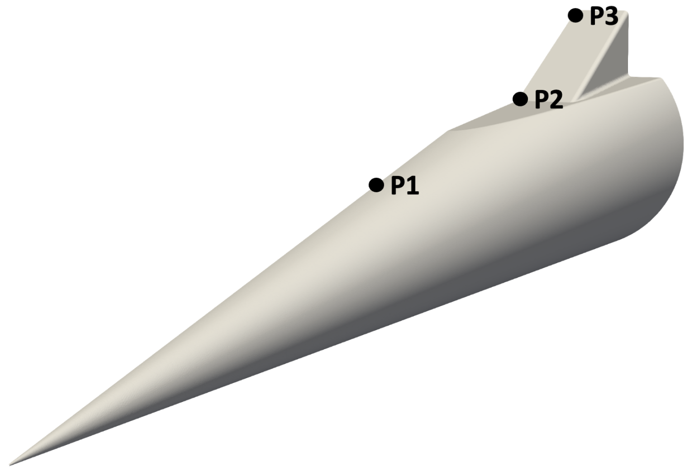

Fig. 5 Geometry used for present computations with location of the three probes marked.

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B. Residuals Algorithm and Separation Bubble Topology

It is well-known [33] that residuals of a time-accurate simulation are directly linked to a decay of the least-stable global flow eigenmode, the damping rate of which can be computed from the logarithmic derivative of the time signal,

$$\omega_1 \equiv \frac{d}{dt}\Big[\ln \left(|\boldsymbol{q}(\boldsymbol{x}, t) - \boldsymbol{q}(\boldsymbol{x})| \right) \Big] \tag{1}$$

where $\boldsymbol{q}(\boldsymbol{x}, t)$ is any flow variable at some particular location (x, y, z) in the flow field at time t, $\boldsymbol{q}(\boldsymbol{x})$ is the final (time independent) value of the chosen flow variable at said location, thus $\boldsymbol{q}(\boldsymbol{x}, t) - \boldsymbol{q}(\boldsymbol{x})$ being the residual, and ω_1 is the amplification rate of the global eigenmode (therefore negative for globally stable flows). Herein, a term residuals algorithm is used to describe the method of recovering amplification rate and amplitude functions of the least-damped global flow eigenmode directly from a numerical simulation. With the decay rate of the least-stable global eigenmode computed, it is possible to recover the corresponding amplitude functions using the linear stability analysis Antsaz as follows: any flow field, within the linear decay region, $\boldsymbol{q} = [\rho, u_x, u_y, u_z, T]^T$ can be decomposed into a steady-state, $\bar{\boldsymbol{q}}(\boldsymbol{x})$, and an infinitesimally small, $\epsilon \ll 1$, perturbation, $\epsilon \boldsymbol{q}'(\boldsymbol{x}, t)$, such that

$$\boldsymbol{q}(\boldsymbol{x}, t) = \bar{\boldsymbol{q}}(\boldsymbol{x}) + \epsilon \boldsymbol{q}'(\boldsymbol{x}, t) \tag{2}$$

where for purpose of the residuals algorithm

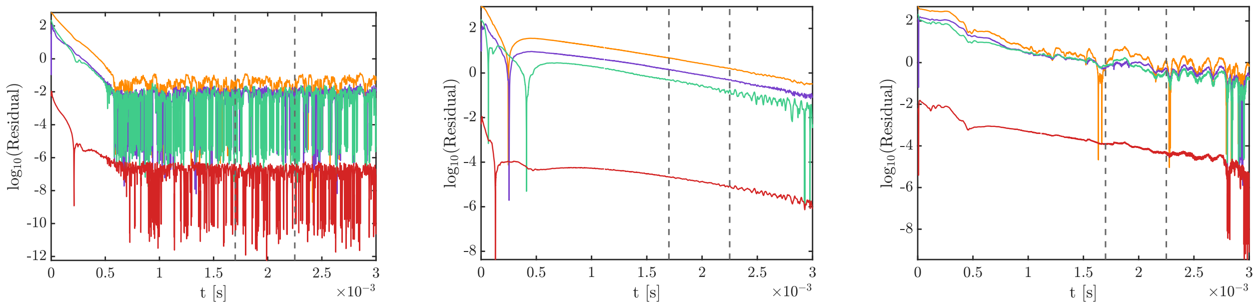
$$\boldsymbol{q}'(\boldsymbol{x}, t) = \hat{\boldsymbol{q}}(\boldsymbol{x}) e^{-\omega_1 t} \tag{3}$$

with $\hat{\boldsymbol{q}}(x, y)$ being the amplitude function that describes the shape of a damped flow disturbance. Equation (3) assumes a stationary mode but the Ansatz can be easily generalized for traveling modes [37]. The damping rate pertains to the decaying mode and remains constant throughout the linear decay. Therefore, should two times be selected in the linear decay regime, a 2-by-2 linear system can be formulated, from which the amplitude functions can be calculated by simple algebraic operations,

$$\epsilon \hat{\boldsymbol{q}}(\boldsymbol{x}_0) = \frac{\boldsymbol{q}'(\boldsymbol{x}_0, t_1) - \boldsymbol{q}'(\boldsymbol{x}_0, t_2)}{e^{-\omega_1 t_1} - e^{-\omega_1 t_2}}, \tag{4}$$

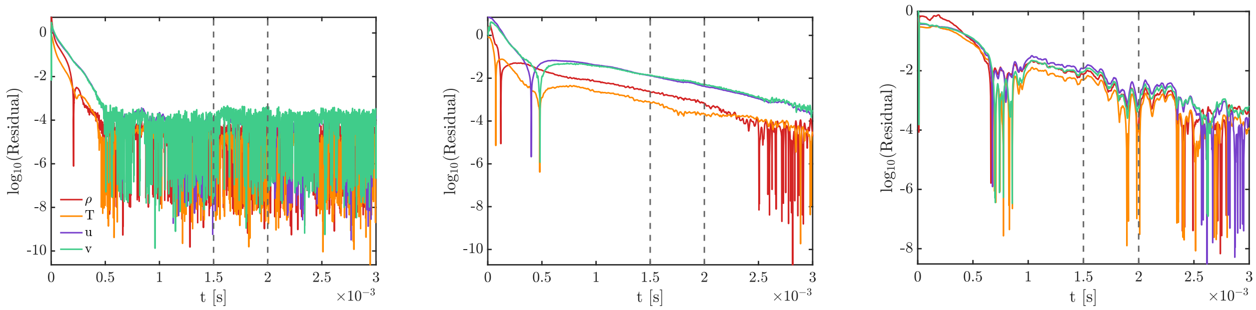
where t_1 and t_2 are the two times within the linear decay regime at which the field are extracted.

In order to analyse the decaying linear perturbations within the numerical simulations performed, the focus is placed on first 0.003 s when flow exhibits linear behavior within the laminar separation bubble. Fig. 6 and Fig. 7 show flow residuals computed at the three probes considered using meshes 1 and 2 respectively. Probe 1, being within the boundary layer upstream of the slice-ramp section, decays and saturates at 10^{-4}, for mesh 2, relatively early in the simulation and is consistent for all flow quantities suggesting, alongside with low-frequency modulation of the saturated signal, that the steady-state has not been reached. In fact, analysis of the residuals from the established flow, see Fig. 8, shows that at chosen set of flow conditions steady-state does not exist and the flow exhibits low frequency oscillations at all probes considered, indicating existence of nonlinear interactions throughout the flow. Probe 2, which is located within the



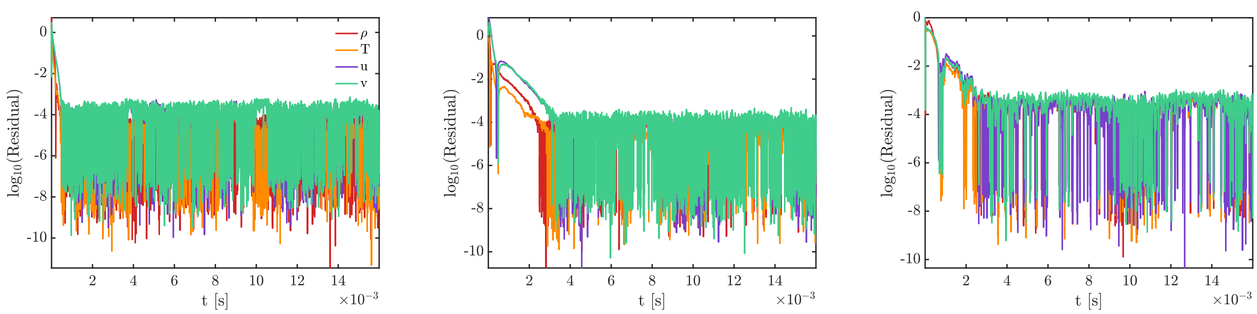
(a) Probe 1 **(b) Probe 2** **(c) Probe 3**

Fig. 6 Self-normalised residuals of flow properties at the three probes considered up to t = 0.003 s computed using mesh 1. Dashed lines represent times used for extracting the least-stable global mode through residuals algorithm.



(a) Probe 1 **(b) Probe 2** **(c) Probe 3**

Fig. 7 Self-normalised residuals of flow properties at the three probes considered up to t = 0.003 s computed using mesh 2. Dashed lines represent times used for extracting the least-stable global mode through residuals algorithm.



(a) Probe 1 **(b) Probe 2** **(c) Probe 3**

Fig. 8 Self-normalised residuals of flow properties at the three probes considered computed using mesh 2 nonlinear nature of the established flow.

separation bubble, undergoes linear decay of the residuals before entering the nonlinear regime as the flow approaches t = 0.003 s mark. Probe 3 in Fig. 6 and Fig. 7, especially for mesh 2, shows lack of the linear regime suggesting that the nonlinearities on the ramp onset at very early times in the simulation.

Using probe 2 as an indicator, the two time-steps considered for residuals algorithm have been extracted at t_1 = 0.0017 s and t_2 = 0.00225 s for mesh 1 and at t_1 = 0.0015 s and t_2 = 0.002 s for mesh 2. Computed least stable perturbations of stream-wise velocity and density along the symmetry plane of the cone-slice-ramp are shown in Fig. 9. Noticeably, the large scale structures of the perturbations computed from present simulations resemble the ones obtained

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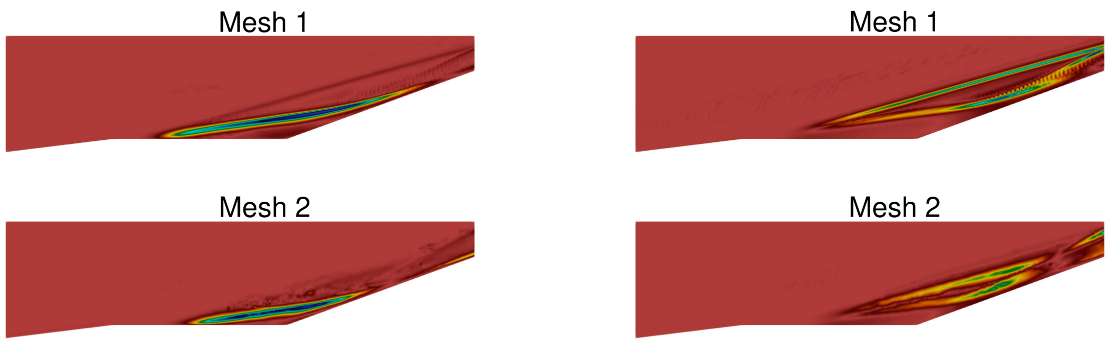
for a supersonic compression corner at $M_\infty = 3$ by Quintanilha [38] who also used residuals algorithm and BiGlobal stability code, LiGHT [39], showing excellent agreement between the two methods. Despite both simulations capturing the large scale structures of the linear perturbation, its stream-wise extent in the mesh 2 is smaller than for the the other simulation. While the perturbation obtained from mesh 2 appears to have irregularities related to the insufficient mesh resolution, grid 1 shows point-wise oscillations within the computed perturbation. Such differences between liner perturbations within direct numerical simulations necessitate further investigations into the most appropriate computational setup. Outlook for future work is discussed in Section V.



(a) abs($\hat{u}_x$)/max($\hat{u}_x$) **(b) abs($\hat{\rho}$)/max($\hat{\rho}$)**

Fig. 9 Self-normalised least stable perturbations extracted from residuals algorithm with the two computational setups considered. Cuts are taken along the symmetry line over the slice-ramp portion of the geometry. Flow is from left to right.

Fig. 10 shows wall-shear lines for the two computations at the final time-step, i.e. in the non-linearly saturated regime. Here, classification of topological structures is qualitative and follows work of Délery [40]. The most significant difference between the two computational strategies is the stream-wise location of the separation node which might arise on account of the unsteadiness in the reattachment zone due to very strong recirculation reaching over 20% of the freestream flow velocity. Past this point, the wall-shear patters are very similar, with the critical point on the ramp for mesh 2 being slightly shifted downstream. Overall, the flow topology maintains the same key features on both meshes. Different locations of the wall-shear nodes for the two configurations indicates different size of the laminar separation bubble, consistently with conclusions reached from the residuals algorithm. Table 5 quantifies the separation, x_S, and reattachment, x_r, locations on the symmetry plane of the model alongside with maximum recirculation strength quantified as percentage of the freestream velocity.

Table 5 Properties of the separated region.

	x_s [m]	x_r [m]	$\max(-u)/u_\infty$ [%]
Mesh 1	0.2765	0.3793	21.25
Mesh 2	0.2888	0.3739	23.43

IV. Computational and Experimental Comparisons

A. Shear Layer

From the computational results, the shear layer height was calculated on the symmetry plane for both simulations using stagnation enthalpy criterion, i.e. boundary layer edge is detected at the height where $h_0/h_{0,\infty} = 0.99$ holds with h_0 and $h_{0,\infty}$ being local and freestream stagnation enthalpy respectively. Computational boundary layer edges have been superimposed on the mean-flow schlieren visualization in Fig. 11. Clear differences between the shear layers from the two meshes are consistent with previously identified (see Section III.B) discrepancies in laminar separation bubble size.

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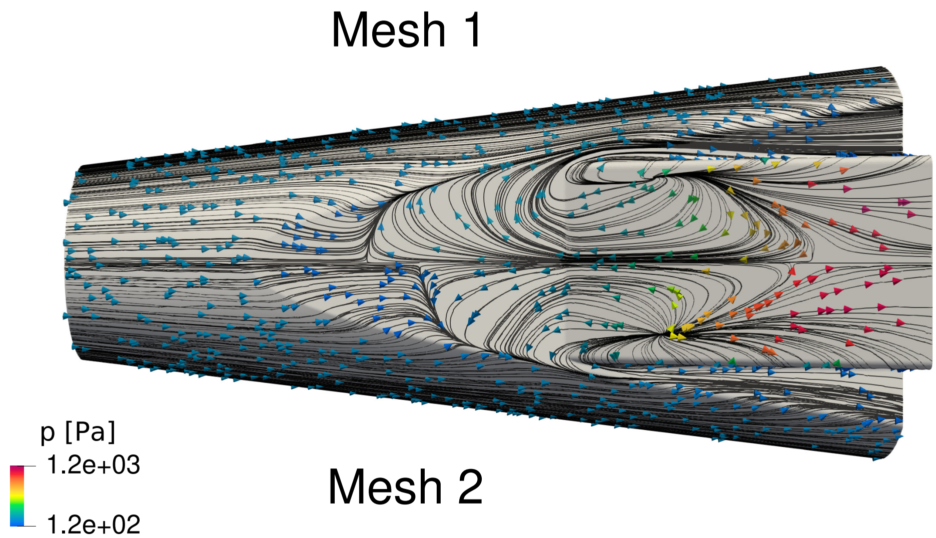

Fig. 10 Wall-shear lines coloured by wall pressure for the two meshes considered in this work. Flow is from left to right.

While the shear layer computed from mesh 1 does closely match the shear layer density gradient in the schlieren, mesh 2 under-predicts the extent of the separation suggesting the former computational strategy is more suitable for detecting the extent of off-surface separation.

B. Surface Measurements

1. Heat Transfer

Experimental and computational surface measurements of heat transfer were non-dimensionalized with Stanton number (C_H) through the relation:

$$C_H = \frac{q_w}{\mu_\infty Re_\infty c_p (T_o - T_w)} \tag{5}$$

where q_w is the heat transfer at the wall, μ_∞ is the freestream viscosity, c_p is the specific heat of air (assumed

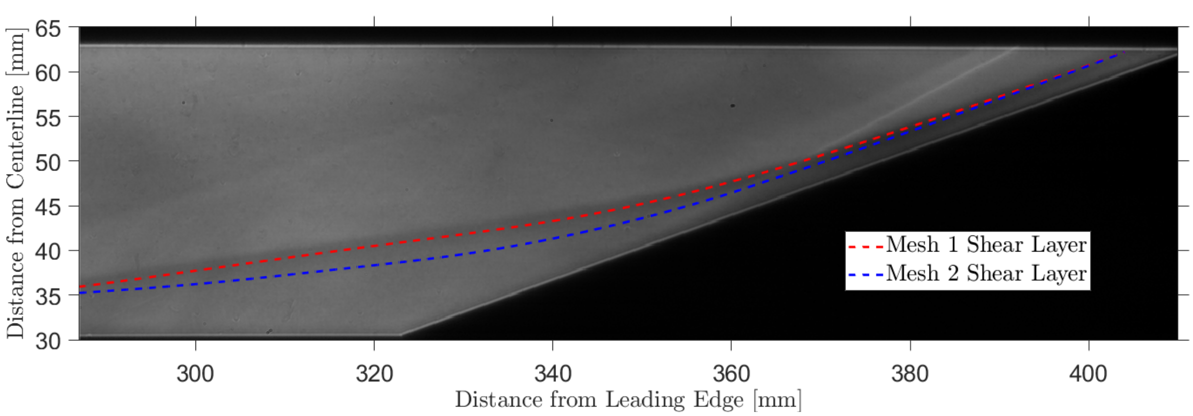

Fig. 11 Time-averaged experimental schlieren and computed shear-layer results from Mesh 1 and Mesh 2.

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constant at 1.009 kJ/kg/K), T_o is the facility stagnation temperature, and T_w is the wall temperature. The Stanton number is the scaled by $\sqrt{Re_\infty}$, which has been found to collapse the heat transfer data onto the laminar result for an insulated flat plate, since the laminar skin friction coefficient $c_f = 0.664/\sqrt{Re_\infty}$ and $C_H \propto c_f$ for a constant Prandtl number as per the Reynolds analogy [3]. The below laminar-scaled heat transfer representation then has units of $m^{-1/2}$.

$$C_{H,laminar} = C_H\sqrt{Re_\infty} \tag{6}$$

Comparison between the heat flux measurements in computations obtained on mesh 2 and in the BAM6QT quiet-flow experimental results are shown in Fig. 12. Slight asymmetries present in the experimental heat transfer are a result of a residual sideslip angle of the model, as discussed in Reference [21]. Despite significant effort placed towards zeroing the combined angles of attack and sideslip of the geometry, surface heat transfer with laminar separation bubbles appear to be particularly sensitive to very small sideslip angles. In the computed case, though the computations assumed half-symmetry, the result is mirrored about the XY plane to provide a side-by-side comparison of ramp heat transfer. The differences in span of the spanwise contours are due to the curvature in the mesh in computations, as only the flat portion is plotted here. These results show good qualitative agreement in general trends of increased heating and in approximate streak origin. However, there are notable differences in the predicted flow structures, particularly in the streaks from edge effects. This could be in part a result of the curvature of the edges in the ramp in computations, as outlined in Section III.A, while in experiment the edges are right angles.

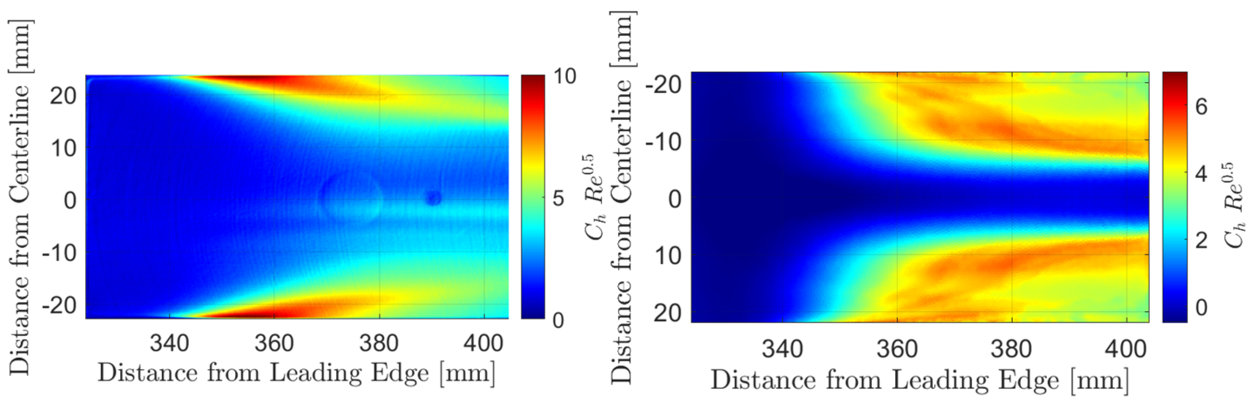

Fig. 12 Laminar-scaled Stanton number on the 20° ramp from experiment (left) and computation (right). The computational result is mirrored about the model centerline. Flow is from left to right.

Fig. 13 shows the laminar-scaled Stanton number plotted along the axial length of the ramp at a 3-mm offset from the model centerline (13a) and across the span of the ramp at 360 mm, 380 mm, and 400 mm from the model leading edge (13b, 13c, and 13d, respectively). The ±3-mm offsets are chosen to avoid a sensor port at 390 mm used for image registration in the experiments. In addition, these profiles make the result of the impact of the residual sideslip angle clear. The inflections in the experimental results in Fig. 13a at 369 mm and 378 mm from the model leading edge are IR camera sensor reflections on the silicon windows. The near-centerline heat transfer for the computations and experiment follow very similar trends, though a heat transfer offset may indicate a difference in predicted physics within the separation bubble or be a result of a fixed wall temperature in the computations that might not reflect the experimental setup accurately. Spanwise heating contours overall show the computations predict edge streaks to trend more quickly towards the centerline of the ramp than measured in experiment. Again, this could be due to the differences in the shape of the edges of the ramps. The overall laminar-scaled heat transfer magnitudes in these spanwise slices are comparable.

2. Centerline Surface Pressure

A comparison of centerline surface pressure between computation on mesh 2 and experiment is plotted in Fig. 14. As discussed, the static pressure presented may have an incorrect offset due to uncertainty in Kulite temperature calibrations, though the trends are correct. Generally, despite possible uncertainties in the calibration, results indicate similar magnitudes of surface static pressure to those found from computation. In the separation bubble, measured by sensors at the 329 mm and 342 mm stations, static pressure trends agree well. The experimental results show a clear

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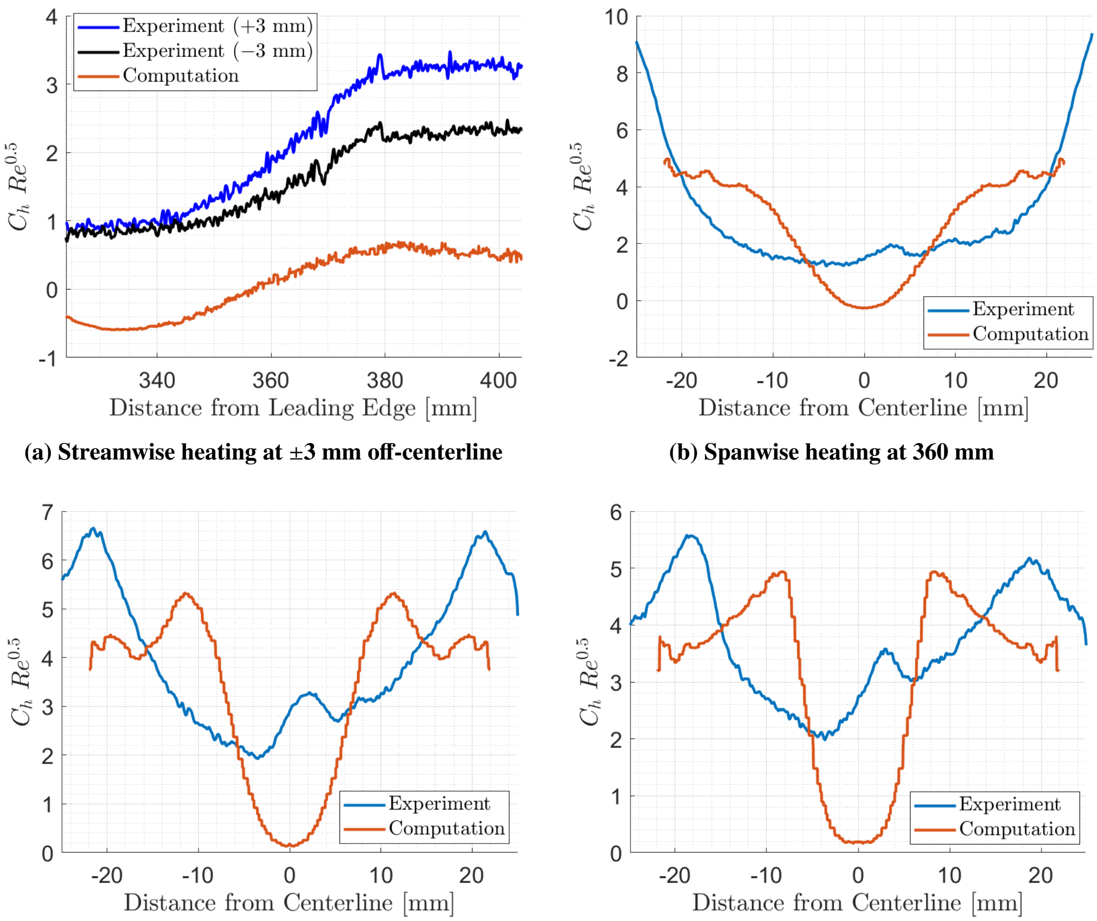

(a) Streamwise heating at ±3 mm off-centerline

(b) Spanwise heating at 360 mm

(c) Spanwise heating at 380 mm

(d) Spanwise heating at 400 mm

Fig. 13 Streamwise laminar-scaled heat transfer at ±3 mm from the model centerline (top left) and three spanwise heat transfer slices at 360 mm (top right), 380 mm (bottom left), and 400 mm (bottom right) from experimental (blue) and computational (orange) results. Near-centerline results include measurements taken on either side of the centerline (blue and black).

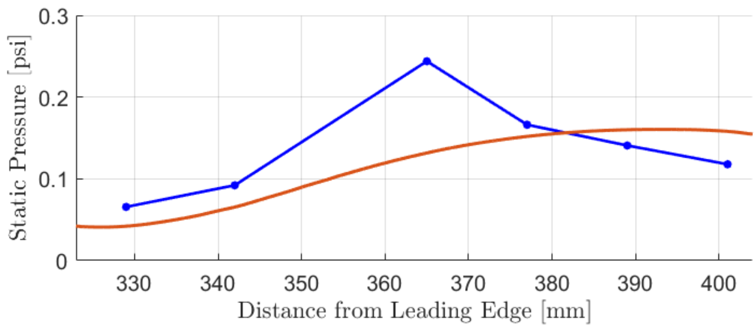

Fig. 14 Centerline static pressure on 20° ramp from Kulites in the BAM6QT (blue) and in computation (orange).

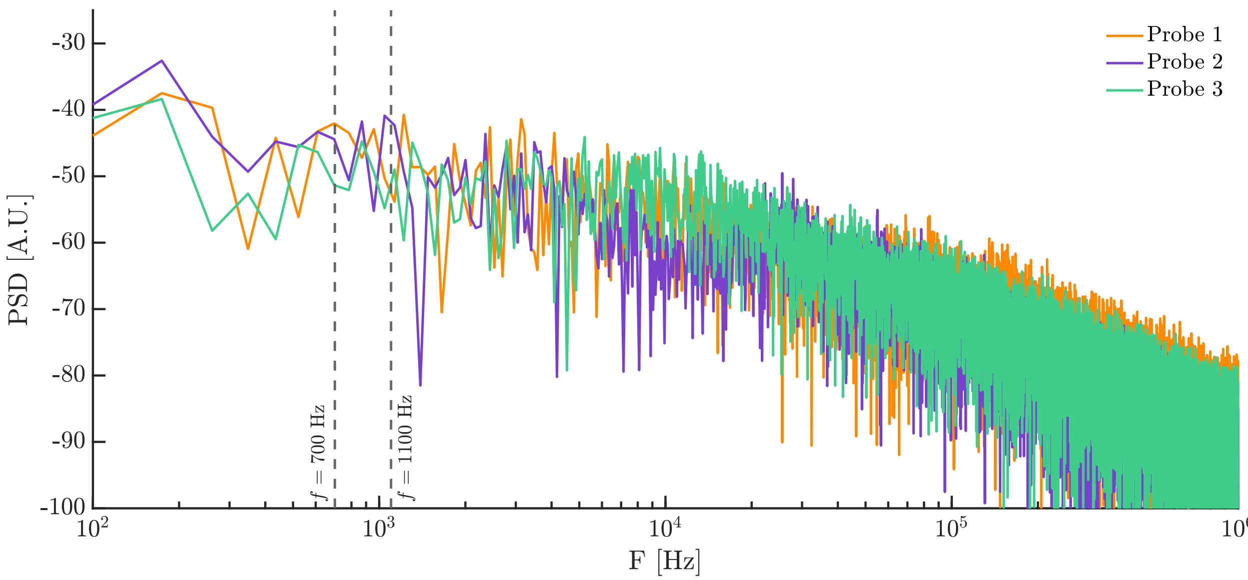

Fig. 15 Power spectral density computed on the temporarily resolved stream-wise velocity data from the three probes. Dashed lines represent frequency regime relevant for this investigation.

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peak near where the shear layer appears to reattach based on schlieren visualization. The computational results predict a more steady rise in static pressure with a peak at near 391 mm from the model leading edge.

C. Time-Resolved Spectral Analysis

Residuals of the full time-resolved data series from the three probes, obtained with mesh 2, are shown in Fig. 8. It becomes immediately clear that a steady-state does not exist for this geometry at these conditions. Instead, the flow reaches a saturated nonlinear stage where low frequency modulation of the temporal signal can be easily noticed. The low-frequency oscillations, unlike the high-frequency ones, can be associated with physical mechanism due to frequency being much smaller than characteristic frequency anticipated for point-wise oscillations based on the mesh cell size. To gain further insight into the nonlinear regime, A Fourier transformation has been applied to those temporarily resolved signals. Power spectral densities shown in Fig. 15, obtained with sampling frequency of approximately 122 MHz, reveal low-frequency energy with a peak in amplitude around 700–1100 Hz. This frequency regime is likely to be representative of unsteadiness in the shear layer and laminar separation bubble.

In order to compare these measured frequencies to experiment, pixels in the high-speed schlieren were chosen in equivalent locations to probes 2 and 3 to compare frequency content. Probe 1 was neglected due to lack of visualization that far upstream on the cone. One additional pixel was chosen in the separated shear layer at an axial position of 365 mm to align with a Kulite at that location. The vertical position of this pixel was selected as the maximum of the derivative of pixel intensity to maximize fluctuation amplitude.

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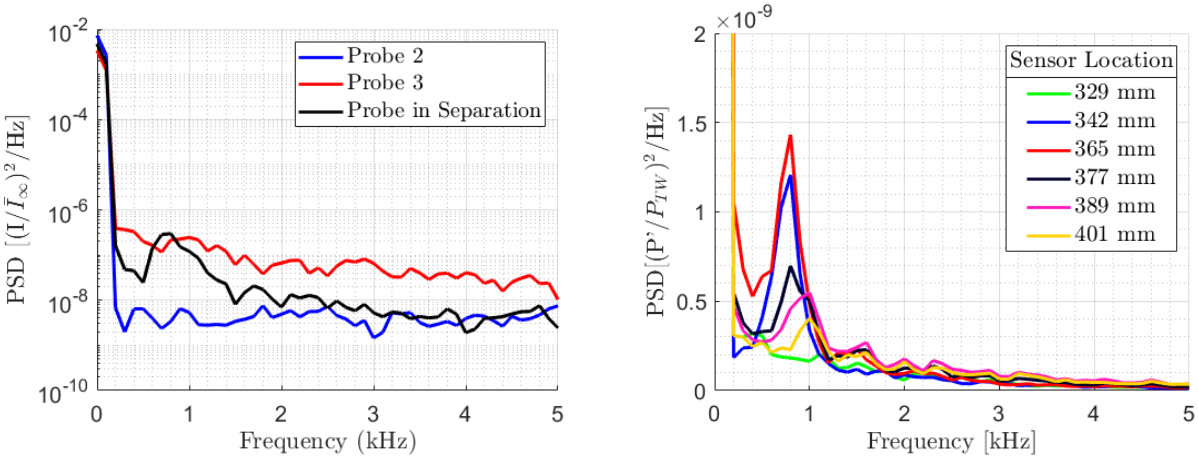

(a) PSDs computed from schlieren probes.

(b) PSDs computed from ramp Kulites.

Fig. 16 Left: PSDs computed from Probe 2 (blue) and Probe 3 (red) in high-speed schlieren. An additional Probe was stationed in the separated shear layer to highlight low-frequency fluctuations (black). Right: Low-frequency centerline pressure fluctuations measured from Kulites.

PSDs were then computed from the time traces of pixel intensity over a 0.05 second interval and are plotted in Fig. 16a. Schlieren probe 2, located near the slice within the separation bubble, shows low broadband frequency content. This is expected from schlieren visualization, since there are no sharp density gradients in the separated flow. Schlieren probe 3, located at the edge of the shear layer downstream of reattachment, reveals higher broadband frequency content, but no significant peaks as seen in probe 3 from the computational temporally-resolved stream-wise velocity data. Some higher-energy peaks are visible around 800–1100 Hz, but it is challenging to distinguish this from the broadband energy. The final probe, selected in the separated shear layer, shows a significant peak in the 700–800 Hz regime, aligning well with frequencies detected in Fig. 15 from the computational results.

Finally, the power spectra was computed from Kulite transducers along the centerline of the ramp, as seen in Fig. 16b. The most-upstream sensor at 329 mm measures low broadband frequency content, indicating the effect of separation bubble fluctuations measured in the schlieren probe in the separated shear layer does not propagate this far into the separated region. By the 342 mm station, significant low-frequency peaks manifest in the 700–800 Hz regime. This peak increases in amplitude by the 365 mm sensor by about 20%. Further downstream, the amplitude of the measured peaks decrease substantially. In addition, the measured peak frequency appears to increase to about 900–1100 Hz by the trailing sensor at 401 mm. All measured frequencies are within ranges predicted by the stream-wise velocity data from the numerical investigation.

V. Summary and Outlook

Comparison of experimental and the present preliminary computational findings reveals varying degrees of agreement concerning heat transfer, surface pressure, and fluctuations. Qualitatively, there is a good agreement in the trends of heat transfer predicted on the 20° ramp, while some discrepancies emerge in prediction of streak location. Schlieren compared to boundary layer thickness extracted from the symmetry plane shows the extent of the separation is well estimated by computation on mesh 1 but not mesh 2, necessitating further investigation. Future computational plans involve systematic investigation of lower Reynolds number flows over the same geometry to fully converge, spatially and temporarily, steady-state solutions, ahead of performing TriGlobal stability analysis to provide insight the into onset of global instabilities and transition mechanisms over this complex geometry.

On the experimental front, centerline-mounted pressure instrumentation precluded the measurement of streaks imparting significant heating that originated on the edges of the ramp. Future experiments will include larger arrays of pressure instrumentation to capture more of the flow physics in these off-centerline regions. Moreover, schlieren was unable to detect substantial density gradients within the region of separated flow, and fluctuations could not be observed. The focus of future experiments will shift towards the implementation of a focusing schlieren apparatus which may improve the sensitivity of the system.

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